

Dr. Jagnyashini Debadarshini

Email : jagnyashini@cs.iitr.ac.in

Mobile : +91-8917403521

Site: <https://jagnyashini.github.io/>

PRESENT AFFILIATION

I am currently working as an **Assistant Professor at the Indian Institute of Technology (IIT) Roorkee**. Prior to this, I worked as a Postdoctoral Research Fellow at National University of Singapore (NUS) with Prof. Mohan Gurusamy. I completed my PhD as a **PMRF** Research Scholar from Indian Institute of Technology (IIT) Bhubaneswar.

RESEARCH INTEREST

My research focuses on the design and development of time- and power-efficient communication protocols for low-power resource constrained distributed systems, such as, *Internet-of-Things* (IoT) and *Wireless Sensor Networks* (WSN). I am particularly interested in developing lightweight edge intelligence frameworks, energy-aware communication protocols, and adaptive networking solutions that enable reliable and real-time data processing under strict power and latency constraints. My work also explores distributed resource management, intelligent network orchestration, and the integration of IoT with emerging paradigms such as edge computing and digital twins. I aim to contribute to next-generation IoT architectures that deliver high performance, scalability, and energy efficiency for practical deployment.

EDUCATION

- **Indian Institute of Technology (IIT), Bhubaneswar** Odisha, India
PhD (PMRF) in Computer Science and Engineering; CGPA : 9.64/10 (Aug 2020 - Apr 2024)
- **Indian Institute of Technology (IIT), Bhubaneswar** Odisha, India
MTech in Computer Science and Engineering; CGPA : 9.56/10 (Jul 2018 - Jun 2020)
- **National Institute of Technology (NIT), Rourkela** Odisha, India
BTech in Computer Science and Engineering; CGPA : 9.22/10 (Jul 2007 - Jun 2011)
- **D.A.V Public School, Unit-8, Bhubaneswar** Odisha, India
Higher Secondary (12); First Division with 86.8% of marks (Jul 2006)
- **D.A.V Public School, Unit-8, Bhubaneswar** Odisha, India
Secondary (10); First Division with 93.4% of marks (Jul 2004)

WORK AND TEACHING EXPERIENCE

- **Asstt. Professor Grade II** (Jun 2025 - Till date)
Indian Institute of Technology (IIT), Roorkee Uttarakhand, India
- **Postdoctoral Research Fellow** (Oct 2024 - May 2025)
Department of Electrical and Computer Engineering, National University of Singapore Singapore
- **Asstt. Professor Grade II** (Aug 2024 - Sep 2024)
Computer Science and Engineering, KIIT Deemed University, Bhubaneswar Odisha, India
- **Research Intern** (May 2024 - July 2024)
Indian Institute of Technology, Bhubaneswar Odisha, India
- **Senior Research Engineer** (Aug 2011 - Aug 2017)
Centre for Development of Telematics (C-DOT), Govt. of India New Delhi, India
- **Programmer** (Sept 2017 - June 2018)
Indian Institute of Technology (IIT), Bhubaneswar Odisha, India
- **Research Intern under the guidance of Prof. Mun Choon Chan** (July 2019 - Nov 2019)
NUS Centre for Research in Privacy Technologies, National University of Singapore Singapore
- **Research Intern under the guidance of Prof. Ajay Jain** (May 2010 - July 2010)
Indian Institute of Technology (IIT), Kanpur UP, India

JOURNALS

- **FETCH: Fast and Efficient Data Collection for Harmonizing Real and Virtual World:** J. Debadarshini, M. Gurusamy., S. Saha, *IEEE Internet of Things Journal (IoT-J)*, 2026.
- **Structure-Adaptive Many-to-Many Data-Sharing for Internet-of-Things:** J. Debadarshini, M. Kausik H, S. Saha, *IEEE Transactions on Network and Service Management (TNSM)*, 2024.
- **LiVeR: Lightweight Vehicle Detection and Classification in Real-Time:** C. Shekhar, J. Debadarshini, S. Saha. *ACM Transactions on Internet of Things (TIOT)*, 2024. (Accepted for journal presentation in SenSys 2024)
- **Efficient Charging of the Electric-Vehicles through IoT-Assisted Coordination with the Charging-Infrastructure:** J. Debadarshini, M. V. Rejeti, C. N. Bhende, S. Saha. *Elsevier AdHoc Network*, 2024.

- **Real-Time Many-to-Many Data-Sharing in Low-Power Wireless Distributed Systems:** J. Debadarshini, M. Tum-mala, S. Saha, O. Landsiedel, M. C. Chan. *IEEE System Journal (SJ)*, 2023.
- **Fine-grained frequencies meet synchronous transmission:** J. Debadarshini, S. Saha. *Elsevier Computer Communica-tion (COMCOM)*, 2023.

CONFERENCE PUBLICATIONS

- **PiLoT: On Overcoming Compute and Memory Constraints for Distributed Machine Learning at the IoT Edge.:** J. Debadarshini, S. Sourav, L. Singh, and S. S. Kumar. *Proceedings of the IEEE 51st Conference on Local Computer Networks (LCN)*, 2026, Coimbra, Portugal. **(Regular)**
- **ReClub: Real-Time Cluster-Formation for UAV-Assisted Scalable Data-Collection.:** J. Debadarshini, S. Sourav, and M. Gurusamy. *Proceedings of the International Conference on Wireless Communications and Networking Conference (WCNC)*, 2026, Kuala Lumpur, Malaysia. **(Regular)**
- **Detecting Stealthy Attackers Hiding in Edge-IoT Systems by Using Micro-distortions.:** S. Sourav, J. Debadarshini, A. Jindal, and B. Chen. *Edge Intelligence for IoT and Healthcare: Advances and Applications, Transactions on Computer Sys-tems and Networks*, Springer, 2026. **(Book Chapter)**
- **TurTle: Topological Structure Formation in Internet-of-Things in Real-Time.:** J. Debadarshini, S. Saha, and M. Gurusamy. *Proceedings of the International Conference on Vehicular Technology Conference (VTC-2025-Spring)*, 2025, Oslo, Norway. **(Regular)**
- **Efficient Coordination in Internet-of-Things through Fast Many-to-Many Conveyance of Bulk-data:** A. Patel, J. Debadarshini, and S. Saha. In *Proceedings of IEEE Asia-Pacific Conference on Communications (APCC)*, 2024, Bali, Indonesia. **(Regular)**
- **SyncCast: Real-Time Self-Adaptive and Fault-Tolerant All-to-All Data-Sharing in IoT:** J. Debadarshini, and S. Saha. In *Proceedings of IEEE International Conference on Network Protocols (ICNP)*, 2023, Reykjavik, Iceland. **(Regular)**
- **LiteCast: Flexible Scalable and Uniform Local Data-Sharing in Real-Time:** J. Debadarshini, and S. Saha. In *Proceedings of IEEE International Conference on Computer Communication and Networks (ICCCN)*, 2023, Honolulu, USA. **(Regular)**
- **An IoT-Assisted Efficient Framework for Multi-Drone Conveyance System:** J. Debadarshini, and S. Saha. In *Pro-ceedings of IEEE Global Communications Conference (GLOBECOM)*, 2023, Kuala Lumpur, Malaysia. **(Regular)**
- **Efficient Simultaneous Intra-Group One-to-All Dissemination:** J. Debadarshini, S. Saha, O. Landsiedel, and M. C. Chan. In *Proceedings of IEEE International Conference on Local Computer Networks (LCN)*, 2020, Sydney, Australia. **(Regular)**
- **Fine-grained Frequencies for Simultaneous Intra-Group One-to-All Dissemination:** J. Debadarshini, C. Sekhar, S. Saha, In *Proceedings of IEEE International Conference on Mobile Ad-Hoc and Smart Systems (MASS)*, 2020, Delhi, India. **(Regular)**
- **Decentralized Load Management in HAN: An IoT-Assisted Approach:** J. Debadarshini, S. Saha, S. R. Saman-taray. In *Proceedings of IEEE International Conference on Smart Grid Communications (SmartGridComm)*, 2022, Singa-pore. **(Regular)**
- **iCharge: An IoT-Assisted Framework for Efficient Charging of the Electric-Vehicles:** J. Debadarshini, M. V. Rejeti, S. Saha and C. N. Bhende. In *Proceedings of IEEE International Conference on COMmunication Systems & NETworks (COMSNETS)*, 2023, India. **(Regular)**
- **Simultaneous Intra-Group Communication: Understanding the Problem Space:** J. Debadarshini, S. Saha. In *Proceedings of IEEE International Conference on Computer Communications (INFOCOM)*, 2022, London, UK. **(Poster)**
- **Efficient Coordination among Electrical Vehicles: An IoT-Assisted Approach:** J. Debadarshini, S. Saha. In *Proceedings of IEEE International Conference on Computer Communications (INFOCOM)*, 2022, London, UK. **(Poster)**
- **Collaborative Load Management in Smart Home Area Network:** J. Debadarshini, S. Saha. In *Proceedings of IEEE International Conference on Distributed Computing Systems (ICDCS)*, 2022, Bologna, Italy. **(Poster)**
- **Divide, Conquer and Merge for Internet-of-Things:** J. Debadarshini, S. Saha. In *Proceedings of IEEE International Conference on Distributed Computing in Sensor Systems (DCOSS)*, 2022, Marina Del Rey, LA, California, USA. **(Poster)**
- **Divide-and-Conquer for Internet-of-Things: Discrete vs Continuous Groups:** J. Debadarshini, and S. Saha. In *Proceedings of IEEE International Conference on COMmunication Systems & NETworks (COMSNETS)*, 2024, India. **(Poster)**
- **Divide-and-Conquer for Internet-of-Things: Discrete vs Continuous Groups:** J. Debadarshini, and S. Saha. In *Proceedings of IEEE International Conference on COMmunication Systems & NETworks (COMSNETS)*, 2024, India. **(Poster)**
- **BLE-based Many-to-Many Data-Sharing for Internet-of-Things:** J. Debadarshini, and S. Saha. In *Proceedings of IEEE International Conference on COMmunication Systems & NETworks (COMSNETS)*, 2024, India. **(Workshop)**
- **BLE-based Many-to-Many Data-Sharing for Internet-of-Things:** J. Debadarshini, and S. Saha. In *Proceedings of IEEE International Conference on COMmunication Systems & NETworks (COMSNETS)*, 2024, India. **(Workshop)**
- **ZoneSync: Real-Time Identification of Zones in IoT-Edge:** M. Kaushik, J. Debadarshini, H. Goyal, S. Saha. In *Proceedings of IEEE International Conference on COMmunication Systems & NETworks (COMSNETS)*, 2023, India. **(Work-shop)**

- **A Lightweight IoT-Based Framework for Vehicular Ad Hoc Network (VANET):** C. Shekhar, J. Debadarshini, P. K. Singh, S. Saha. *In Proceedings of IEEE International Conference on COMmunication Systems & NETworkS (COMSNETS)*, 2023, India. (*Workshop*)

PROJECT ASSIGNMENT

- **Design and Development of PCRF and SGW module in Long Term Evaluation (LTE-4G) Cellular Network:** Worked on developing and testing software modules like Serving Gateway(SGW), Policy Charging and Control Entity (PCRF). The software required designing, debugging and testing of communication protocols DIAMETER, GTP-U, and GTP-C.
- **Intelligent Network (IN) Services in C-DOT Cellular Softswitch:** Worked on developing and testing software modules like Service Switching Point, Service Control Point, and Media Server for IN services. The software required designing, debugging and testing of communication protocols INAP, TCAP, SCCP, M3UA, SCTP, ISUP, and SIP.

HONORS AND AWARDS

- Received **Prime Minister Research Fellowship (PMRF), Govt. of India** from July 2020 to July 2024, for pursuing **PhD** at Indian Institute of Technology (IIT) Bhubaneswar, India.
- Received **scholarship from Ministry of Human Resource Development (MHRD), Govt. of India**, from July 2018 to July 2020, for pursuing **MTech** at Indian Institute of Technology(IIT) Bhubaneswar, India.
- Ranked **2nd in MTech**, in the department of Computer Science and Engineering, Indian Institute of Technology (IIT) Bhubaneswar, India, in 2020.
- Received the **Best employee award** at C-DOT, Govt. of India, in 2016.
- Ranked **4th in BTech**, in the department of Computer Science and Engineering, National Institute of Technology (NIT) Rourkela, India, in 2011.
- Qualified with a **GATE Percentile of 99.65** in 2014.
- Selected in the **ten-week SURGE** (Summer Under Graduate Research Grant for Excellence) program at Indian Institute of Technology (IIT) Kanpur, in 2010.
- Ranked **4th in School** in Class 10th with **Merit Certification** from CBSE for the highest mark in Science and Sanskrit, in 2004.